

3. First bud production

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for survival analysis.

```
FB_df <-

# Load the raw data frame.
df %>%

# Define the follow-up time for each individual:
# if the hydra produced a first bud -> time of first bud production (event),
# if the hydra died -> time of death (censor),
# if the hydra did not produce a first bud and stayed alive
# -> 42 days (no event, censor at the end of the experiment).
mutate(age_followup_days = case_when(
  FB_state == 1 ~ age_FB,
  FB_state == 0 & death_state == 1 ~ age_death,
  TRUE ~ 42),
```

```

age_followup = age_followup_days / 7,
FB_event = FB_state) %>%

#' Build a competing-risk variable for the cumulative incidence functions:
#' if the hydra did not produce a first bud and stayed alive -> 0 (no event),
#' if the hydra produced a first bud -> 1 (event),
#' if the hydra died -> 2 (competing risk of death).
mutate(status_competing = factor(case_when(FB_event == 1 ~ 1,
                                           death_state == 1 ~ 2,
                                           TRUE ~ 0))) %>%

#' Build the variable tum_state_early: an individual is considered
#' tumoral only if the tumor appeared before or at the same time as the
#' supernumerary tentacle event.
mutate(age_tum_weeks = age_tum / 7,
       tum_state_early = case_when(tum_state == "Tumoral" &
                                   !is.na(age_tum_weeks) &
                                   age_tum_weeks <= age_followup ~ "Tumoral",
                                   TRUE ~ "Healthy"),
       tum_state_early = factor(tum_state_early, levels = c("Healthy", "Tumoral"))) %>%

#' Remove missing data.
filter(!is.na(age_followup)) %>%

#' Keep only the variables needed for the analysis.
select(age_followup, FB_event, status_competing, treatment_condition, lineage,
       tum_state_early, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

FB_df$lineage <- relevel(factor(FB_df$lineage), ref = "HO_MT")
FB_df$treatment_condition <- relevel(factor(FB_df$treatment_condition), ref = "Control")

```

3 Survival object

Create a survival object with follow-up time and event indicator.

```

FB_surv_object <- Surv(FB_df$age_followup, FB_df$FB_event)

```

4 Proportional hazards assumption check : Schoenfeld residuals

Fit a preliminary full Cox model (fixed effects only).

```

FB_cox <- coxph(Surv(age_followup, FB_event) ~ treatment_condition * lineage *
               tum_state_early, data = FB_df)

```

Test the proportional hazards assumption.

```
print(cox.zph(FB_cox))
```

```
##                chisq df      p
## treatment_condition      5.55  1  0.019
## lineage                131.26  5 < 2e-16
## tum_state_early          4.39  1  0.036
## treatment_condition:lineage    55.59  5 9.9e-11
## treatment_condition:tum_state_early    6.19  1  0.013
## lineage:tum_state_early    5.44  2  0.066
## treatment_condition:lineage:tum_state_early    2.50  1  0.114
## GLOBAL                176.70 16 < 2e-16
```

The proportional hazards assumption is violated for at least one term, an accelerated failure time mixed-effects model will be used.

5 AIC comparison of different distributions

Compare intercept-only accelerated failure time models fitted with different parametric distributions to identify the best-fitting one.

```
FB_dist_weib <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "weibull")

FB_dist_lnorm <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                           dist = "lognormal")

FB_dist_llog <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                           dist = "loglogistic")

FB_dist_exp <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "exponential")

AIC(FB_dist_weib, FB_dist_lnorm, FB_dist_llog, FB_dist_exp) %>% arrange(AIC)
```

```
##          df      AIC
## FB_dist_lnorm  2 2521.574
## FB_dist_llog   2 2530.357
## FB_dist_weib   2 2634.431
## FB_dist_exp    1 2648.340
```

The model with a log-normal distribution has the lowest AIC, we'll use it for the models.

6 Model selection

6.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

FB_R_0 <- SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage * tum_state_early,
  data = FB_df, dist = "lognormal")

FB_R_1 <- SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage * tum_state_early +
  (1 | parent_unique),
  data = FB_df, dist = "lognormal")

FB_R_2 <- SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage * tum_state_early +
  (1 | replicate_unique),
  data = FB_df, dist = "lognormal")

FB_R_3 <- SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage * tum_state_early +
  (1 | parent_unique) + (1 | replicate_unique),
  data = FB_df, dist = "lognormal")

AIC(FB_R_0, FB_R_1, FB_R_2, FB_R_3) %>% arrange(AIC)

```

```

##      df      AIC
## FB_R_2 26 2041.720
## FB_R_3 27 2043.720
## FB_R_0 25 2197.253
## FB_R_1 26 2199.253

```

The model with the replicate id as a random effect (FB_R_2) has the lowest AIC, this random effect structure is therefore selected.

6.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```

FB_F_0 <- SurvregME(Surv(age_followup, FB_event) ~ 1 + (1 | replicate_unique), data =
FB_df, dist = "lognormal")
FB_F_1 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_2 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + (1 | replicate_unique), data
= FB_df, dist = "lognormal")
FB_F_3 <- SurvregME(Surv(age_followup, FB_event) ~ tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_4 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition + lineage + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_5 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition + tum_state_early
+ (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_6 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_7 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition * lineage + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_8 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition * tum_state_early
+ (1 | replicate_unique), data = FB_df, dist = "lognormal")

```

```

FB_F_9 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_10 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition + lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_11 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition * lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_12 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition * tum_state_early
+ lineage + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_13 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early +
treatment_condition + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_14 <- SurvregME(Surv(age_followup, FB_event) ~ treatment_condition * lineage *
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")

AIC(FB_F_0, FB_F_1, FB_F_2, FB_F_3, FB_F_4, FB_F_5, FB_F_6, FB_F_7,
    FB_F_8, FB_F_9, FB_F_10, FB_F_11, FB_F_12, FB_F_13, FB_F_14) %>% arrange(AIC)

```

```

##      df      AIC
## FB_F_11 15 2022.674
## FB_F_12 11 2039.850
## FB_F_10 10 2040.381
## FB_F_14 26 2041.720
## FB_F_6   9 2044.891
## FB_F_13 15 2047.613
## FB_F_9   14 2051.542
## FB_F_7   14 2114.692
## FB_F_4    9 2123.404
## FB_F_2    8 2132.611
## FB_F_8    6 2207.268
## FB_F_5    5 2209.009
## FB_F_3    4 2214.100
## FB_F_1    4 2266.732
## FB_F_0    3 2275.753

```

The model with the interaction between the treatment condition and the lineage, and with the additive effect of tumoral state has the lowest AIC and is therefore selected (FB_F_11).

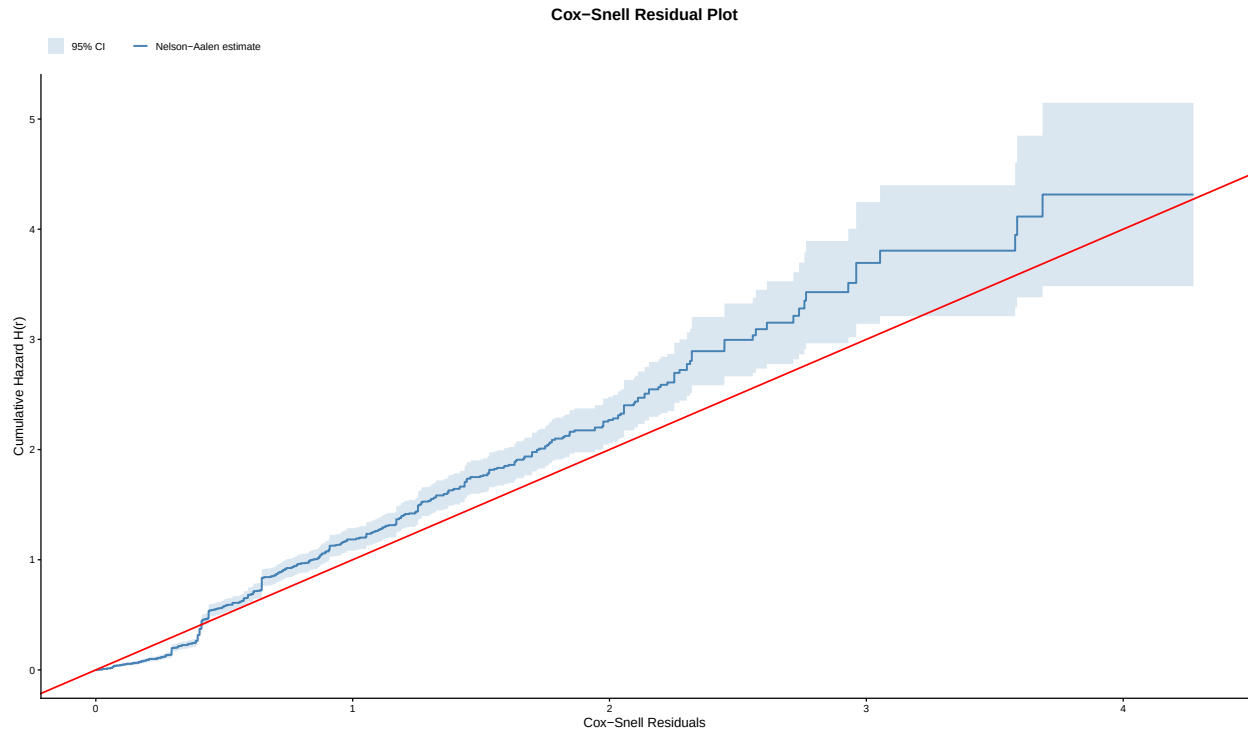
6.3 Goodness-of-fit evaluation

Assess goodness-of-fit of the selected model using Cox-Snell residuals.

```

plot_cox_snell_survregme(model = FB_F_11, data = FB_df, time_col = "age_followup",
event_col = "FB_event")

```



The curve follows a 45 degree exponential, we validate the fit of the selected model.

6.4 Selected model summary

#' Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage + tum_state_early + (1 |
  replicate_unique),
  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##   tum_state_early + (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##
```

	Estimate	Std. Error	z value
## treatment_conditionShaking treatment	0.048981	0.170051	0.2880
## lineageHC_MT	2.211147	0.298500	7.4075
## lineageHO_SPC	0.185987	0.257831	0.7214
## lineageHO_SPT	1.474533	0.266992	5.5228

```

## lineageHO_VLN                2.069535    0.252228    8.2050
## lineageHV_GAL                1.971254    0.267067    7.3811
## tum_state_earlyTumoral       2.234789    0.234979    9.5106
## treatment_conditionShaking treatment:lineageHC_MT    0.578781    0.269384    2.1485
## treatment_conditionShaking treatment:lineageHO_SPC -0.036154    0.238204   -0.1518
## treatment_conditionShaking treatment:lineageHO_SPT -0.516544    0.242203   -2.1327
## treatment_conditionShaking treatment:lineageHO_VLN    0.547600    0.225972    2.4233
## treatment_conditionShaking treatment:lineageHV_GAL    0.212865    0.239548    0.8886
##                               Pr(>|z|)
## treatment_conditionShaking treatment                0.77332
## lineageHC_MT                                         1.287e-13 ***
## lineageHO_SPC                                       0.47069
## lineageHO_SPT                                       3.337e-08 ***
## lineageHO_VLN                                       2.306e-16 ***
## lineageHV_GAL                                       1.570e-13 ***
## tum_state_earlyTumoral                             < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHC_MT    0.03167 *
## treatment_conditionShaking treatment:lineageHO_SPC    0.87936
## treatment_conditionShaking treatment:lineageHO_SPT    0.03295 *
## treatment_conditionShaking treatment:lineageHO_VLN    0.01538 *
## treatment_conditionShaking treatment:lineageHV_GAL    0.37421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

```

```

FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPC")
summary(SurvregME(Surv(age_followup, FB_event) ~
                  treatment_condition * lineage + tum_state_early + (1 |
replicate_unique),
                  data = FB_df, dist = "lognormal"))

```

```

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##   tum_state_early + (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##

```

```

##                                     Estimate Std. Error z value
## treatment_conditionShaking treatment      0.012831    0.167869  0.0764
## lineageHO_MT                        -0.185979    0.257831 -0.7213
## lineageHC_MT                        2.025164    0.296385  6.8329
## lineageHO_SPT                       1.288545    0.265838  4.8471
## lineageHO_VLN                       1.883548    0.249845  7.5389
## lineageHV_GAL                       1.785271    0.264853  6.7406
## tum_state_earlyTumoral              2.234794    0.234979  9.5106
## treatment_conditionShaking treatment:lineageHO_MT  0.036143    0.238204  0.1517
## treatment_conditionShaking treatment:lineageHC_MT  0.614928    0.267964  2.2948
## treatment_conditionShaking treatment:lineageHO_SPT -0.480394    0.240519 -1.9973
## treatment_conditionShaking treatment:lineageHO_VLN  0.583752    0.224262  2.6030
## treatment_conditionShaking treatment:lineageHV_GAL  0.249012    0.237980  1.0464
##                                     Pr(>|z|)
## treatment_conditionShaking treatment      0.939071
## lineageHO_MT                        0.470712
## lineageHC_MT                        8.322e-12 ***
## lineageHO_SPT                       1.253e-06 ***
## lineageHO_VLN                       4.741e-14 ***
## lineageHV_GAL                       1.577e-11 ***
## tum_state_earlyTumoral              < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHO_MT  0.879400
## treatment_conditionShaking treatment:lineageHC_MT  0.021744 *
## treatment_conditionShaking treatment:lineageHO_SPT  0.045790 *
## treatment_conditionShaking treatment:lineageHO_VLN  0.009242 **
## treatment_conditionShaking treatment:lineageHV_GAL  0.295396
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

```

```

FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPT")
summary(SurvregME(Surv(age_followup, FB_event) ~
                  treatment_condition * lineage + tum_state_early + (1 |
replicate_unique),
                  data = FB_df, dist = "lognormal"))

```

```

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##      tum_state_early + (1 | replicate_unique)
##

```

```
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## treatment_conditionShaking treatment -0.46757    0.17350 -2.6949
## lineageHO_SPC -1.28855    0.26584 -4.8471
## lineageHO_MT -1.47453    0.26699 -5.5228
## lineageHC_MT  0.73661    0.29238  2.5194
## lineageHO_VLN  0.59499    0.24624  2.4163
## lineageHV_GAL  0.49671    0.26189  1.8966
## tum_state_earlyTumoral 2.23480    0.23498  9.5106
## treatment_conditionShaking treatment:lineageHO_SPC 0.48040    0.24052  1.9974
## treatment_conditionShaking treatment:lineageHO_MT  0.51654    0.24220  2.1327
## treatment_conditionShaking treatment:lineageHC_MT  1.09533    0.27264  4.0176
## treatment_conditionShaking treatment:lineageHO_VLN 1.06416    0.22974  4.6319
## treatment_conditionShaking treatment:lineageHV_GAL 0.72942    0.24241  3.0090
##                                     Pr(>|z|)
## treatment_conditionShaking treatment 0.007041 **
## lineageHO_SPC 1.253e-06 ***
## lineageHO_MT 3.337e-08 ***
## lineageHC_MT 0.011756 *
## lineageHO_VLN 0.015678 *
## lineageHV_GAL 0.057877 .
## tum_state_earlyTumoral < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHO_SPC 0.045787 *
## treatment_conditionShaking treatment:lineageHO_MT 0.032950 *
## treatment_conditionShaking treatment:lineageHC_MT 5.880e-05 ***
## treatment_conditionShaking treatment:lineageHO_VLN 3.623e-06 ***
## treatment_conditionShaking treatment:lineageHV_GAL 0.002621 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)
```

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_VLN")
summary(SurvregME(Surv(age_followup, FB_event) ~
                  treatment_condition * lineage + tum_state_early + (1 |
replicate_unique),
                  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
```

```
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##   tum_state_early + (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##               Estimate Std. Error z value
## treatment_conditionShaking treatment      0.596595    0.149017    4.0035
## lineageHO_SPT                               -0.594993    0.246240   -2.4163
## lineageHO_SPC                               -1.883540    0.249845   -7.5388
## lineageHO_MT                                -2.069524    0.252228   -8.2050
## lineageHC_MT                                 0.141629    0.272716    0.5193
## lineageHV_GAL                               -0.098264    0.240411   -0.4087
## tum_state_earlyTumoral                      2.234804    0.234979    9.5107
## treatment_conditionShaking treatment:lineageHO_SPT -1.064162    0.229745   -4.6319
## treatment_conditionShaking treatment:lineageHO_SPC -0.583767    0.224262   -2.6031
## treatment_conditionShaking treatment:lineageHO_MT  -0.547622    0.225972   -2.4234
## treatment_conditionShaking treatment:lineageHC_MT   0.031167    0.255372    0.1220
## treatment_conditionShaking treatment:lineageHV_GAL -0.334754    0.224607   -1.4904
##
##               Pr(>|z|)
## treatment_conditionShaking treatment      6.240e-05 ***
## lineageHO_SPT                               0.01568 *
## lineageHO_SPC                               4.742e-14 ***
## lineageHO_MT                                2.307e-16 ***
## lineageHC_MT                                 0.60353
## lineageHV_GAL                               0.68273
## tum_state_earlyTumoral                      < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHO_SPT 3.623e-06 ***
## treatment_conditionShaking treatment:lineageHO_SPC  0.00924 **
## treatment_conditionShaking treatment:lineageHO_MT   0.01538 *
## treatment_conditionShaking treatment:lineageHC_MT   0.90286
## treatment_conditionShaking treatment:lineageHV_GAL  0.13612
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)
```

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HC_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage + tum_state_early + (1 |
  replicate_unique),
```

```
data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##     tum_state_early + (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## treatment_conditionShaking treatment      0.627755   0.209080   3.0025
## lineageHO_VLN                        -0.141603   0.272716  -0.5192
## lineageHO_SPT                        -0.736598   0.292376  -2.5194
## lineageHO_SPC                        -2.025149   0.296384  -6.8329
## lineageHO_MT                         -2.211131   0.298500  -7.4075
## lineageHV_GAL                       -0.239892   0.287121  -0.8355
## tum_state_earlyTumoral                2.234790   0.234979   9.5106
## treatment_conditionShaking treatment:lineageHO_VLN -0.031172   0.255372  -0.1221
## treatment_conditionShaking treatment:lineageHO_SPT -1.095320   0.272637  -4.0175
## treatment_conditionShaking treatment:lineageHO_SPC -0.614927   0.267964  -2.2948
## treatment_conditionShaking treatment:lineageHO_MT  -0.578778   0.269384  -2.1485
## treatment_conditionShaking treatment:lineageHV_GAL -0.365902   0.268223  -1.3642
##                                     Pr(>|z|)
## treatment_conditionShaking treatment      0.002678 **
## lineageHO_VLN                        0.603600
## lineageHO_SPT                        0.011757 *
## lineageHO_SPC                        8.324e-12 ***
## lineageHO_MT                         1.287e-13 ***
## lineageHV_GAL                        0.403433
## tum_state_earlyTumoral                < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHO_VLN  0.902847
## treatment_conditionShaking treatment:lineageHO_SPT  5.882e-05 ***
## treatment_conditionShaking treatment:lineageHO_SPC  0.021744 *
## treatment_conditionShaking treatment:lineageHO_MT   0.031672 *
## treatment_conditionShaking treatment:lineageHV_GAL  0.172514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)
```

```

FB_df$lineage <- relevel(FB_df$lineage, ref = "HV_GAL")
summary(SurvregME(Surv(age_followup, FB_event) ~
  treatment_condition * lineage + tum_state_early + (1 |
replicate_unique),
  data = FB_df, dist = "lognormal"))

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ treatment_condition * lineage +
##   tum_state_early + (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## treatment_conditionShaking treatment      0.261860    0.168784  1.5515
## lineageHC_MT                               0.239910    0.287122  0.8356
## lineageHO_VLN                             0.098294    0.240411  0.4089
## lineageHO_SPT                            -0.496709    0.261894 -1.8966
## lineageHO_SPC                           -1.785257    0.264853 -6.7406
## lineageHO_MT                            -1.971238    0.267067 -7.3811
## tum_state_earlyTumoral                    2.234796    0.234979  9.5106
## treatment_conditionShaking treatment:lineageHC_MT  0.365898    0.268223  1.3642
## treatment_conditionShaking treatment:lineageHO_VLN  0.334720    0.224607  1.4902
## treatment_conditionShaking treatment:lineageHO_SPT -0.729426    0.242415 -3.0090
## treatment_conditionShaking treatment:lineageHO_SPC -0.249033    0.237980 -1.0464
## treatment_conditionShaking treatment:lineageHO_MT  -0.212888    0.239548 -0.8887
##                                     Pr(>|z|)
## treatment_conditionShaking treatment      0.120793
## lineageHC_MT                               0.403398
## lineageHO_VLN                             0.682642
## lineageHO_SPT                             0.057880 .
## lineageHO_SPC                            1.578e-11 ***
## lineageHO_MT                             1.570e-13 ***
## tum_state_earlyTumoral                    < 2.2e-16 ***
## treatment_conditionShaking treatment:lineageHC_MT  0.172519
## treatment_conditionShaking treatment:lineageHO_VLN  0.136159
## treatment_conditionShaking treatment:lineageHO_SPT  0.002621 **
## treatment_conditionShaking treatment:lineageHO_SPC  0.295356
## treatment_conditionShaking treatment:lineageHO_MT  0.374160
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)

```

```
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)
```

7 Median times

7.1 By treatment condition

Estimate cumulative incidence functions for the first bud event, accounting for the competing risk of death, stratified by treatment condition.

```
FB_cuminc_treatment <- cuminc(Surv(age_followup, status_competing) ~ treatment_condition,
data = FB_df)
```

For each stratum, extract the median time to event and its 95CI.

```
FB_medians_treatment <- tidy(FB_cuminc_treatment) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_treatment)
```

```
## # A tibble: 2 x 4
##   strata      median_time ci_low ci_high
##   <fct>          <dbl>   <dbl>   <dbl>
## 1 Control          1.14     1.14     1.14
## 2 Shaking treatment 1.14     1.14     1.29
```

7.2 By lineage

Same method as above, but stratified by lineage.

```
FB_cuminc_lineage <- cuminc(Surv(age_followup, status_competing) ~ lineage, data = FB_df)

FB_medians_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
```

```

    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_lineage)

```

```

## # A tibble: 6 x 4
##   strata median_time ci_low ci_high
##   <fct>      <dbl> <dbl>  <dbl>
## 1 HV_GAL      1.71  1.29    2
## 2 HC_MT        3    2.43    3
## 3 HO_VLN       1     1    2.43
## 4 HO_SPT      1.29  1.14    1.29
## 5 HO_SPC      0.286 0.286    1.14
## 6 HO_MT       0.286 0.286    1

```

7.3 By treatment condition and lineage

Same method as above, but stratified by treatment condition and lineage.

```

FB_cuminc_interaction <- cuminc(Surv(age_followup, status_competing) ~
treatment_condition + lineage, data = FB_df)

FB_medians_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_interaction)

```

```

## # A tibble: 12 x 4
##   strata                median_time ci_low ci_high
##   <fct>                <dbl> <dbl>  <dbl>
## 1 Control, HV_GAL      1.71  1.71    2
## 2 Shaking treatment, HV_GAL 1.71  1    2.43
## 3 Control, HC_MT       1.43  1.14    3
## 4 Shaking treatment, HC_MT  3     3     3
## 5 Control, HO_VLN       1     1    1.14
## 6 Shaking treatment, HO_VLN 1.14  1    3.14
## 7 Control, HO_SPT      1.29  1.14    1.57
## 8 Shaking treatment, HO_SPT 1.14  1.14    1.29

```

```
## 9 Control, HO_SPC          0.286 0.286 1.14
## 10 Shaking treatment, HO_SPC 0.286 0.286 1.14
## 11 Control, HO_MT          0.286 0.286 1
## 12 Shaking treatment, HO_MT 0.286 0.286 1
```

7.4 By tumoral state

Same method as above, but stratified by tumoral state.

```
FB_cuminc_tum <- cuminc(Surv(age_followup, status_competing) ~ tum_state_early, data =
FB_df)

FB_medians_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_tum)
```

```
## # A tibble: 2 x 4
##   strata median_time ci_low ci_high
##   <fct>      <dbl> <dbl> <dbl>
## 1 Healthy      1.14  1.14  1.14
## 2 Tumoral       3    2.29  NA
```

8 Cumulative incidences

8.1 By week and treatment condition

Convert the continuous cumulative incidence function into a weekly summary table: for each stratum of treatment condition and each week, show the estimate with its 95CI.

```
FB_incidence_treatment <- tidy(FB_cuminc_treatment) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_treatment)
```

```
## # A tibble: 14 x 5
##   strata      week estimate_cuminc conf.low conf.high
##   <fct>    <dbl>         <dbl>    <dbl>    <dbl>
## 1 Control      0             0      NA      NA
## 2 Control      1         0.433    0.386    0.479
## 3 Control      2         0.736    0.692    0.775
## 4 Control      3         0.894    0.860    0.919
## 5 Control      4         0.907    0.876    0.931
## 6 Control      5         0.931    0.902    0.951
## 7 Control      6         0.947    0.921    0.964
## 8 Shaking treatment 0             0      NA      NA
## 9 Shaking treatment 1         0.421    0.374    0.467
## 10 Shaking treatment 2         0.650    0.603    0.693
## 11 Shaking treatment 3         0.789    0.748    0.825
## 12 Shaking treatment 4         0.822    0.782    0.855
## 13 Shaking treatment 5         0.880    0.845    0.907
## 14 Shaking treatment 6         0.882    0.848    0.909
```

8.2 By week and lineage

Same method as above, but stratified by lineage.

```
FB_incidence_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_lineage)
```

```
## # A tibble: 42 x 5
##   strata week estimate_cuminc conf.low conf.high
##   <fct> <dbl>         <dbl>    <dbl>    <dbl>
## 1 HV_GAL 0             0      NA      NA
## 2 HV_GAL 1         0.319    0.245    0.396
## 3 HV_GAL 2         0.597    0.512    0.672
## 4 HV_GAL 3         0.868    0.800    0.914
## 5 HV_GAL 4         0.875    0.808    0.920
## 6 HV_GAL 5         0.931    0.874    0.962
## 7 HV_GAL 6         0.938    0.882    0.967
## 8 HC_MT 0             0      NA      NA
## 9 HC_MT 1         0.0729    0.0320    0.137
## 10 HC_MT 2         0.375    0.279    0.471
## 11 HC_MT 3         0.844    0.754    0.903
## 12 HC_MT 4         0.917    0.838    0.958
## 13 HC_MT 5         0.938    0.862    0.972
## 14 HC_MT 6         0.938    0.862    0.972
## 15 HO_VLN 0             0      NA      NA
## 16 HO_VLN 1         0.536    0.463    0.604
```

## 17	HO_VLN	2	0.573	0.500	0.640
## 18	HO_VLN	3	0.672	0.600	0.734
## 19	HO_VLN	4	0.703	0.633	0.763
## 20	HO_VLN	5	0.813	0.749	0.861
## 21	HO_VLN	6	0.849	0.789	0.893
## 22	HO_SPT	0	0	NA	NA
## 23	HO_SPT	1	0.215	0.152	0.286
## 24	HO_SPT	2	0.819	0.746	0.874
## 25	HO_SPT	3	0.868	0.800	0.914
## 26	HO_SPT	4	0.882	0.816	0.925
## 27	HO_SPT	5	0.882	0.816	0.925
## 28	HO_SPT	6	0.882	0.816	0.925
## 29	HO_SPC	0	0	NA	NA
## 30	HO_SPC	1	0.549	0.463	0.626
## 31	HO_SPC	2	0.882	0.816	0.925
## 32	HO_SPC	3	0.924	0.865	0.957
## 33	HO_SPC	4	0.951	0.899	0.977
## 34	HO_SPC	5	0.979	0.931	0.994
## 35	HO_SPC	6	0.979	0.931	0.994
## 36	HO_MT	0	0	NA	NA
## 37	HO_MT	1	0.715	0.634	0.781
## 38	HO_MT	2	0.847	0.778	0.896
## 39	HO_MT	3	0.931	0.874	0.962
## 40	HO_MT	4	0.931	0.874	0.962
## 41	HO_MT	5	0.931	0.874	0.962
## 42	HO_MT	6	0.931	0.874	0.962

8.3 By week, treatment condition, and lineage

Same method as above, but stratified by treatment condition and lineage.

```
FB_incidence_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_interaction)
```

```
## # A tibble: 84 x 5
##   strata                week estimate_cuminc conf.low conf.high
##   <fct>                <dbl>         <dbl>    <dbl>    <dbl>
## 1 Control, HV_GAL      0             0      NA      NA
## 2 Control, HV_GAL      1          0.222    0.134    0.324
## 3 Control, HV_GAL      2          0.639    0.515    0.739
## 4 Control, HV_GAL      3          0.986    0.898    0.998
## 5 Control, HV_GAL      4             NA      NA      NA
## 6 Control, HV_GAL      5             NA      NA      NA
## 7 Control, HV_GAL      6             NA      NA      NA
```

## 8	Shaking treatment, HV_GAL	0	0	NA	NA
## 9	Shaking treatment, HV_GAL	1	0.417	0.302	0.528
## 10	Shaking treatment, HV_GAL	2	0.556	0.432	0.662
## 11	Shaking treatment, HV_GAL	3	0.75	0.631	0.835
## 12	Shaking treatment, HV_GAL	4	0.75	0.631	0.835
## 13	Shaking treatment, HV_GAL	5	0.861	0.755	0.924
## 14	Shaking treatment, HV_GAL	6	0.875	0.771	0.934
## 15	Control, HC_MT	0	0	NA	NA
## 16	Control, HC_MT	1	0.0833	0.0263	0.183
## 17	Control, HC_MT	2	0.521	0.370	0.652
## 18	Control, HC_MT	3	0.917	0.788	0.969
## 19	Control, HC_MT	4	0.979	0.777	0.998
## 20	Control, HC_MT	5	0.979	0.777	0.998
## 21	Control, HC_MT	6	0.979	0.777	0.998
## 22	Shaking treatment, HC_MT	0	0	NA	NA
## 23	Shaking treatment, HC_MT	1	0.0625	0.0160	0.156
## 24	Shaking treatment, HC_MT	2	0.229	0.122	0.356
## 25	Shaking treatment, HC_MT	3	0.771	0.623	0.867
## 26	Shaking treatment, HC_MT	4	0.854	0.713	0.929
## 27	Shaking treatment, HC_MT	5	0.896	0.759	0.957
## 28	Shaking treatment, HC_MT	6	0.896	0.759	0.957
## 29	Control, HO_VLN	0	0	NA	NA
## 30	Control, HO_VLN	1	0.604	0.499	0.694
## 31	Control, HO_VLN	2	0.635	0.530	0.723
## 32	Control, HO_VLN	3	0.75	0.650	0.825
## 33	Control, HO_VLN	4	0.771	0.672	0.843
## 34	Control, HO_VLN	5	0.865	0.776	0.920
## 35	Control, HO_VLN	6	0.938	0.863	0.972
## 36	Shaking treatment, HO_VLN	0	0	NA	NA
## 37	Shaking treatment, HO_VLN	1	0.469	0.366	0.565
## 38	Shaking treatment, HO_VLN	2	0.510	0.406	0.606
## 39	Shaking treatment, HO_VLN	3	0.594	0.488	0.685
## 40	Shaking treatment, HO_VLN	4	0.635	0.530	0.723
## 41	Shaking treatment, HO_VLN	5	0.760	0.661	0.835
## 42	Shaking treatment, HO_VLN	6	0.760	0.661	0.835
## 43	Control, HO_SPT	0	0	NA	NA
## 44	Control, HO_SPT	1	0.194	0.112	0.293
## 45	Control, HO_SPT	2	0.819	0.707	0.892
## 46	Control, HO_SPT	3	0.847	0.738	0.914
## 47	Control, HO_SPT	4	0.847	0.738	0.914
## 48	Control, HO_SPT	5	0.847	0.738	0.914
## 49	Control, HO_SPT	6	0.847	0.738	0.914
## 50	Shaking treatment, HO_SPT	0	0	NA	NA
## 51	Shaking treatment, HO_SPT	1	0.236	0.145	0.340
## 52	Shaking treatment, HO_SPT	2	0.819	0.707	0.892
## 53	Shaking treatment, HO_SPT	3	0.889	0.786	0.944
## 54	Shaking treatment, HO_SPT	4	0.917	0.818	0.963
## 55	Shaking treatment, HO_SPT	5	0.917	0.818	0.963
## 56	Shaking treatment, HO_SPT	6	0.917	0.818	0.963
## 57	Control, HO_SPC	0	0	NA	NA
## 58	Control, HO_SPC	1	0.542	0.419	0.649
## 59	Control, HO_SPC	2	0.889	0.787	0.944
## 60	Control, HO_SPC	3	0.972	0.870	0.994
## 61	Control, HO_SPC	4	0.972	0.870	0.994

## 62	Control, HO_SPC	5	NA	NA	NA
## 63	Control, HO_SPC	6	NA	NA	NA
## 64	Shaking treatment, HO_SPC	0	0	NA	NA
## 65	Shaking treatment, HO_SPC	1	0.556	0.433	0.662
## 66	Shaking treatment, HO_SPC	2	0.875	0.771	0.934
## 67	Shaking treatment, HO_SPC	3	0.875	0.771	0.934
## 68	Shaking treatment, HO_SPC	4	0.931	0.836	0.972
## 69	Shaking treatment, HO_SPC	5	0.972	0.874	0.994
## 70	Shaking treatment, HO_SPC	6	0.972	0.874	0.994
## 71	Control, HO_MT	0	0	NA	NA
## 72	Control, HO_MT	1	0.778	0.663	0.858
## 73	Control, HO_MT	2	0.875	0.771	0.934
## 74	Control, HO_MT	3	0.944	0.852	0.980
## 75	Control, HO_MT	4	0.944	0.852	0.980
## 76	Control, HO_MT	5	0.944	0.852	0.980
## 77	Control, HO_MT	6	0.944	0.852	0.980
## 78	Shaking treatment, HO_MT	0	0	NA	NA
## 79	Shaking treatment, HO_MT	1	0.653	0.530	0.751
## 80	Shaking treatment, HO_MT	2	0.819	0.709	0.891
## 81	Shaking treatment, HO_MT	3	0.917	0.821	0.962
## 82	Shaking treatment, HO_MT	4	0.917	0.821	0.962
## 83	Shaking treatment, HO_MT	5	0.917	0.821	0.962
## 84	Shaking treatment, HO_MT	6	0.917	0.821	0.962

8.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```
FB_incidence_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_tum)
```

```
## # A tibble: 14 x 5
##   strata week estimate_cuminc conf.low conf.high
##   <fct>  <dbl>         <dbl>    <dbl>    <dbl>
## 1 Healthy    0             0      NA      NA
## 2 Healthy    1          0.447  0.413  0.481
## 3 Healthy    2          0.712  0.680  0.742
## 4 Healthy    3          0.858  0.832  0.880
## 5 Healthy    4          0.880  0.856  0.900
## 6 Healthy    5          0.919  0.898  0.935
## 7 Healthy    6          0.928  0.908  0.944
## 8 Tumoral    0             0      NA      NA
## 9 Tumoral    1         0.0244 0.00185 0.112
## 10 Tumoral   2          0.317  0.181  0.462
```

```
## 11 Tumoral      3          0.512  0.349      0.654
## 12 Tumoral      4          0.561  0.394      0.699
## 13 Tumoral      5          0.634  0.464      0.763
## 14 Tumoral      6          0.634  0.464      0.763
```

9 Cumulative proportions

9.1 By week and treatment condition

Compute raw cumulative proportions of first bud production per week, stratified by treatment condition.

```
FB_proportions_treatment <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(treatment_condition) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(treatment_condition, week, n, total_N, raw_proportion = p)

print(FB_proportions_treatment)
```

```
## # A tibble: 12 x 5
##   treatment_condition week      n total_N raw_proportion
##   <fct>              <dbl> <int>   <int>         <dbl>
## 1 Control            1    187    432         0.433
## 2 Control            2    318    432         0.736
## 3 Control            3    386    432         0.894
## 4 Control            4    392    432         0.907
## 5 Control            5    402    432         0.931
## 6 Control            6    409    432         0.947
## 7 Shaking treatment  1    182    432         0.421
## 8 Shaking treatment  2    281    432         0.650
## 9 Shaking treatment  3    341    432         0.789
## 10 Shaking treatment 4    355    432         0.822
## 11 Shaking treatment 5    380    432         0.880
## 12 Shaking treatment 6    381    432         0.882
```

9.2 By week and lineage

Same method as above, but stratified by lineage.

```
FB_proportions_lineage <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$",),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_lineage)
```

```
## # A tibble: 36 x 5
##   lineage week   n total_N raw_proportion
##   <fct>   <dbl> <int>   <int>         <dbl>
## 1 HV_GAL     1    46    144         0.319
## 2 HV_GAL     2    86    144         0.597
## 3 HV_GAL     3   125    144         0.868
## 4 HV_GAL     4   126    144         0.875
## 5 HV_GAL     5   134    144         0.931
## 6 HV_GAL     6   135    144         0.938
## 7 HC_MT      1     7     96         0.0729
## 8 HC_MT      2    36     96         0.375
## 9 HC_MT      3    81     96         0.844
## 10 HC_MT     4    88     96         0.917
## 11 HC_MT     5    90     96         0.938
## 12 HC_MT     6    90     96         0.938
## 13 HO_VLN     1   103    192         0.536
## 14 HO_VLN     2   110    192         0.573
## 15 HO_VLN     3   129    192         0.672
## 16 HO_VLN     4   135    192         0.703
## 17 HO_VLN     5   156    192         0.812
## 18 HO_VLN     6   163    192         0.849
## 19 HO_SPT     1    31    144         0.215
## 20 HO_SPT     2   118    144         0.819
## 21 HO_SPT     3   125    144         0.868
## 22 HO_SPT     4   127    144         0.882
```

```
## 23 HO_SPT      5   127   144      0.882
## 24 HO_SPT      6   127   144      0.882
## 25 HO_SPC      1    79   144      0.549
## 26 HO_SPC      2   127   144      0.882
## 27 HO_SPC      3   133   144      0.924
## 28 HO_SPC      4   137   144      0.951
## 29 HO_SPC      5   141   144      0.979
## 30 HO_SPC      6   141   144      0.979
## 31 HO_MT       1   103   144      0.715
## 32 HO_MT       2   122   144      0.847
## 33 HO_MT       3   134   144      0.931
## 34 HO_MT       4   134   144      0.931
## 35 HO_MT       5   134   144      0.931
## 36 HO_MT       6   134   144      0.931
```

9.3 By week, treatment condition, and lineage

Same method as above, but stratified by treatment condition and lineage.

```
FB_proportions_interaction <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(treatment_condition, lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$",
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(treatment_condition, lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_interaction)
```

```
## # A tibble: 72 x 6
##   treatment_condition lineage week    n total_N raw_proportion
##   <fct>                <fct>  <dbl> <int>   <int>         <dbl>
## 1 Control              HV_GAL    1    16     72         0.222
## 2 Control              HV_GAL    2    46     72         0.639
## 3 Control              HV_GAL    3    71     72         0.986
## 4 Control              HV_GAL    4    72     72          1
## 5 Control              HV_GAL    5    72     72          1
## 6 Control              HV_GAL    6    72     72          1
```

## 7 Control	HC_MT	1	4	48	0.0833
## 8 Control	HC_MT	2	25	48	0.521
## 9 Control	HC_MT	3	44	48	0.917
## 10 Control	HC_MT	4	47	48	0.979
## 11 Control	HC_MT	5	47	48	0.979
## 12 Control	HC_MT	6	47	48	0.979
## 13 Control	HO_VLN	1	58	96	0.604
## 14 Control	HO_VLN	2	61	96	0.635
## 15 Control	HO_VLN	3	72	96	0.75
## 16 Control	HO_VLN	4	74	96	0.771
## 17 Control	HO_VLN	5	83	96	0.865
## 18 Control	HO_VLN	6	90	96	0.938
## 19 Control	HO_SPT	1	14	72	0.194
## 20 Control	HO_SPT	2	59	72	0.819
## 21 Control	HO_SPT	3	61	72	0.847
## 22 Control	HO_SPT	4	61	72	0.847
## 23 Control	HO_SPT	5	61	72	0.847
## 24 Control	HO_SPT	6	61	72	0.847
## 25 Control	HO_SPC	1	39	72	0.542
## 26 Control	HO_SPC	2	64	72	0.889
## 27 Control	HO_SPC	3	70	72	0.972
## 28 Control	HO_SPC	4	70	72	0.972
## 29 Control	HO_SPC	5	71	72	0.986
## 30 Control	HO_SPC	6	71	72	0.986
## 31 Control	HO_MT	1	56	72	0.778
## 32 Control	HO_MT	2	63	72	0.875
## 33 Control	HO_MT	3	68	72	0.944
## 34 Control	HO_MT	4	68	72	0.944
## 35 Control	HO_MT	5	68	72	0.944
## 36 Control	HO_MT	6	68	72	0.944
## 37 Shaking treatment	HV_GAL	1	30	72	0.417
## 38 Shaking treatment	HV_GAL	2	40	72	0.556
## 39 Shaking treatment	HV_GAL	3	54	72	0.75
## 40 Shaking treatment	HV_GAL	4	54	72	0.75
## 41 Shaking treatment	HV_GAL	5	62	72	0.861
## 42 Shaking treatment	HV_GAL	6	63	72	0.875
## 43 Shaking treatment	HC_MT	1	3	48	0.0625
## 44 Shaking treatment	HC_MT	2	11	48	0.229
## 45 Shaking treatment	HC_MT	3	37	48	0.771
## 46 Shaking treatment	HC_MT	4	41	48	0.854
## 47 Shaking treatment	HC_MT	5	43	48	0.896
## 48 Shaking treatment	HC_MT	6	43	48	0.896
## 49 Shaking treatment	HO_VLN	1	45	96	0.469
## 50 Shaking treatment	HO_VLN	2	49	96	0.510
## 51 Shaking treatment	HO_VLN	3	57	96	0.594
## 52 Shaking treatment	HO_VLN	4	61	96	0.635
## 53 Shaking treatment	HO_VLN	5	73	96	0.760
## 54 Shaking treatment	HO_VLN	6	73	96	0.760
## 55 Shaking treatment	HO_SPT	1	17	72	0.236
## 56 Shaking treatment	HO_SPT	2	59	72	0.819
## 57 Shaking treatment	HO_SPT	3	64	72	0.889
## 58 Shaking treatment	HO_SPT	4	66	72	0.917
## 59 Shaking treatment	HO_SPT	5	66	72	0.917
## 60 Shaking treatment	HO_SPT	6	66	72	0.917

## 61	Shaking treatment	HO_SPC	1	40	72	0.556
## 62	Shaking treatment	HO_SPC	2	63	72	0.875
## 63	Shaking treatment	HO_SPC	3	63	72	0.875
## 64	Shaking treatment	HO_SPC	4	67	72	0.931
## 65	Shaking treatment	HO_SPC	5	70	72	0.972
## 66	Shaking treatment	HO_SPC	6	70	72	0.972
## 67	Shaking treatment	HO_MT	1	47	72	0.653
## 68	Shaking treatment	HO_MT	2	59	72	0.819
## 69	Shaking treatment	HO_MT	3	66	72	0.917
## 70	Shaking treatment	HO_MT	4	66	72	0.917
## 71	Shaking treatment	HO_MT	5	66	72	0.917
## 72	Shaking treatment	HO_MT	6	66	72	0.917

9.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```
FB_proportions_tum <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(tum_state_early) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$",
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(tum_state_early, week, n, total_N, raw_proportion = p)

print(FB_proportions_tum)
```

```
## # A tibble: 12 x 5
##   tum_state_early week      n total_N raw_proportion
##   <fct>          <dbl> <int>   <int>         <dbl>
## 1 Healthy        1    368    823         0.447
## 2 Healthy        2    586    823         0.712
## 3 Healthy        3    706    823         0.858
## 4 Healthy        4    724    823         0.880
## 5 Healthy        5    756    823         0.919
## 6 Healthy        6    764    823         0.928
## 7 Tumoral        1      1     41         0.0244
## 8 Tumoral        2     13     41         0.317
```

## 9 Tumoral	3	21	41	0.512
## 10 Tumoral	4	23	41	0.561
## 11 Tumoral	5	26	41	0.634
## 12 Tumoral	6	26	41	0.634

10 Plots of cumulative incidence of first bud production

10.1 Common themes

Theme of y-axis, and its presence or absence on the plots

```
scale_y <- scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.25),
                              labels = percent, position = "right")
```

Fix to hide y-axis numbers, ticks, and titles on the right and left

```
hide_y <- theme(axis.text.y.right = element_blank(),
                axis.ticks.y.right = element_blank(),
                axis.title.y.right = element_blank(),
                axis.text.y = element_blank(),
                axis.ticks.y = element_blank(),
                axis.title.y = element_blank())
```

Fix to show y-axis numbers and ticks on the right

```
show_y <- theme(axis.text.y.right = element_text(color = "black", size = 14),
                axis.ticks.y.right = element_line(),
                axis.text.y = element_blank(),
                axis.ticks.y = element_blank(),
                axis.title.y = element_blank())
```

Theme of x-axis, and its presence or absence on the plots.

```
scale_x <- scale_x_continuous(limits = c(0, 6), breaks = seq(0, 6, by = 2))

hide_x <- theme(axis.text.x = element_blank(),
                axis.ticks.x = element_blank(),
                axis.title.x = element_blank())

show_x <- theme(axis.text.x = element_text(color = "black", size = 14),
                axis.ticks.x = element_line())
```

Color scale for tumoral state

```
tum_scale <- list(
  scale_color_manual(values = tum_colors),
  scale_fill_manual(values = tum_colors))
```

Color scales for treatment condition

```
treatment_scale <- list(
  scale_color_manual(values = treatment_condition_colors, breaks = c("Control", "Shaking
treatment")),
  scale_fill_manual(values = treatment_condition_colors, breaks = c("Control", "Shaking
treatment")))
```

Theme shared by panels A, B, and C (angle = 90 for bottom-to-top reading)

```
theme_common <- theme(
  axis.title.x = element_text(color = "black", size = 14, margin = margin(t = 10)),
  axis.title.y.right = element_text(color = "black", size = 14, angle = 90, margin =
margin(l = 15)),
  plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Specific label configurations for precise placement

```
labs_base <- labs(x = NULL, y = NULL, title = NULL)
labs_x_only <- labs(x = "Time (weeks)", y = NULL, title = NULL)
labs_y_only <- labs(x = NULL, y = "Cumulative incidence of\nfirst bud production", title
= NULL)
labs_xy <- labs(x = "Time (weeks)", y = "Cumulative incidence of\nfirst bud
production", title = NULL)
```

10.2 Panel A: interaction between treatment condition and lineage

10.2.1 HO_SPT

```
A1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
  data = FB_df %>% filter(lineage == "HO_SPT") %>%
  mutate(treatment_condition = factor(treatment_condition, levels =
c("Control", "Shaking treatment")))) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + hide_y +
theme(axis.text.x = element_text(color = "black", size = 14),
  legend.position = "none") +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
bold("_SPT")),
  color = "black", size = 5, hjust = 1, vjust = 0)
```

10.2.2 HO_VLN

```
A2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
  data = FB_df %>% filter(lineage == "HO_VLN") %>%
  mutate(treatment_condition = factor(treatment_condition, levels =
c("Control", "Shaking treatment")))) %>%
```

```
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_x_only + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + hide_y +
theme(axis.text.x = element_text(color = "black", size = 14),
       legend.position = "none") +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
bold("_VLN")),
         color = "black", size = 5, hjust = 1, vjust = 0)
```

10.2.3 HC_MT

```
A3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ treatment_condition,
             data = FB_df %>% filter(lineage == "HC_MT") %>%
             mutate(treatment_condition = factor(treatment_condition, levels =
             c("Control", "Shaking treatment")))) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_y_only + scale_y + scale_x + treatment_scale +
theme_bw() + theme_common + show_y +
theme(axis.text.x = element_text(color = "black", size = 14),
       legend.position = "none") +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hc") * bold("_MT")),
         color = "black", size = 5, hjust = 1, vjust = 0)
```

Extract legend from the plot A3

```
shared_treatment_legend <- cowplot::get_legend(
  A3 + theme(legend.position = "bottom",
             legend.direction = "horizontal",
             legend.text = element_text(size = 13),
             legend.title = element_text(size = 14, face = "bold")))
```

Combine A1, A2, and A3 into panel “A”

```
A <- (A1 + A2 + A3 + plot_layout(nrow = 1)) +
plot_annotation(title = "Interaction between the shaking treatment and the lineage",
               theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
               size = 16, margin = margin(b = 10))))
```

10.3 Panel B: by tumoral state

Cumulative incidence of first bud production by tumoral state.

```
B <- cuminc(Surv(age_followup, factor(status_competing)) ~ tum_state_early, data = FB_df)
%>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_xy + scale_y + tum_scale + scale_x +
```

```

theme_bw() + theme_common + show_y +
theme(axis.text.x = element_text(color = "black", size = 14),
      axis.text.y.right = element_text(color = "black", size = 14),
      axis.ticks.y.right = element_line(),
      legend.position = "none") +
plot_annotation(title = "Tumoral state effect",
                theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                size = 16, margin = margin(b = 10))))

```

Extract legend from this panel

```

shared_tumor_legend <- cowplot::get_legend(
  B + theme(legend.position = "bottom",
            legend.direction = "horizontal",
            legend.text = element_text(size = 13),
            legend.title = element_text(size = 14, face = "bold")))

```

10.4 Panel C: by lineage

10.4.1 HO_MT

```

C1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HO_MT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") * bold("_MT")),
           color = "black", size = 5, hjust = 1, vjust = 0)

```

10.4.2 HO_SPC

```

C2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HO_SPC")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
bold("_SPC")),
           color = "black", size = 5, hjust = 1, vjust = 0)

```

10.4.3 HO_SPT

```

C3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HO_SPT")) %>%

```

```
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
theme(axis.text.y.right = element_text(color = "black", size = 14),
axis.ticks.y.right = element_line()) +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
bold("_SPT")),
color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.4 HO_VLN

```
C4 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
data = FB_df %>% filter(lineage == "HO_VLN")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + show_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
bold("_VLN")),
color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.5 HC_MT

```
C5 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
data = FB_df %>% filter(lineage == "HC_MT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_x_only + scale_y + scale_x +
theme_bw() + theme_common + hide_y + show_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hc") * bold("_MT")),
color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.6 HV_GAL

```
C6 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
data = FB_df %>% filter(lineage == "HV_GAL")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_y_only + scale_y + scale_x +
theme_bw() + theme_common + show_y + show_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hv") *
bold("_GAL")),
color = "black", size = 5, hjust = 1, vjust = 0)
```

Create panel C combining plots C1 to C6.

```
C <- (C1 + C2 + C3) / (C4 + C5 + C6) +
  plot_layout(heights = c(1, 1)) +
  plot_annotation(title = "Lineage effect",
                  theme = theme(plot.title = element_text(
                    hjust = 0.5, face = "bold", size = 16, margin = margin(b = 10))))
```

10.5 Assembly

Add a letter tag to the first sub-plot of each panel.

```
A[[1]] <- A[[1]] + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face =
"bold"))
B <- B + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))
C[[1]][[1]] <- C[[1]][[1]] + labs(tag = "C") + theme(plot.tag = element_text(size = 20,
face = "bold"))
```

Convert each panel into a graphical object.

```
A <- patchwork::patchworkGrob(A)
```

Wrap B in patchwork before conversion to force perfect axis alignment with A and C

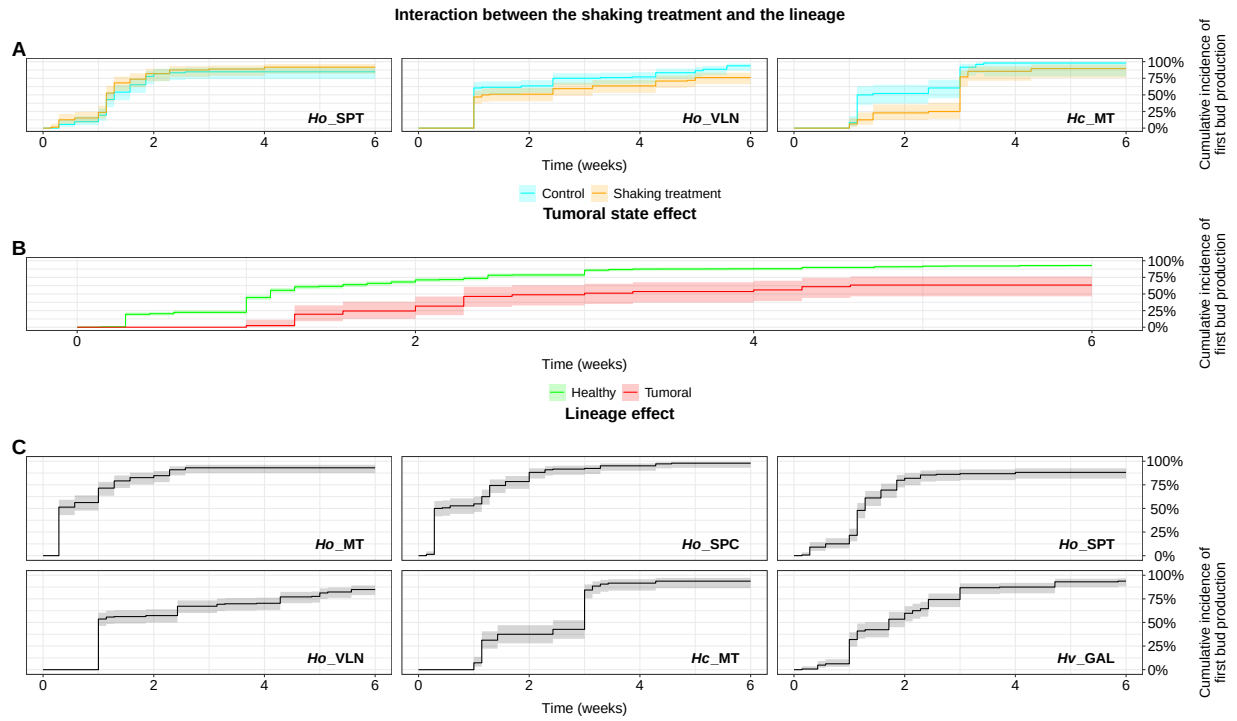
```
B <- patchwork::patchworkGrob(B + patchwork::plot_layout(ncol = 1))
C <- patchwork::patchworkGrob(C)
```

Combine panels and shared legends vertically into a single column, aligning axes.

```
final_FB_panel <- cowplot::plot_grid(A,
                                     shared_treatment_legend,
                                     B,
                                     shared_tumor_legend,
                                     C,
                                     ncol = 1,
                                     align = "v",
                                     axis = "lr",
                                     rel_heights = c(1, 0.1, 1, 0.1, 1.8)) +
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Show the final figure.

```
print(final_FB_panel)
```



```
ggsave("figure_4.png", plot = final_FB_panel, width = 21, height = 29.7, units = "cm",
dpi = 300)
```